

Model Evaluation Report
Started: 2026-04-12 23:10:21
Finished: 2026-04-12 23:33:35

Model Architecture: DenseNet121
Early Stopping Patience: 5
Phase 1 Learning Rate: 0.001
Phase 2 Learning Rate: 1e-06
Phase 1 Epochs: 27 (Limit: 30)
Phase 2 Epochs: 6 (Limit: 50)

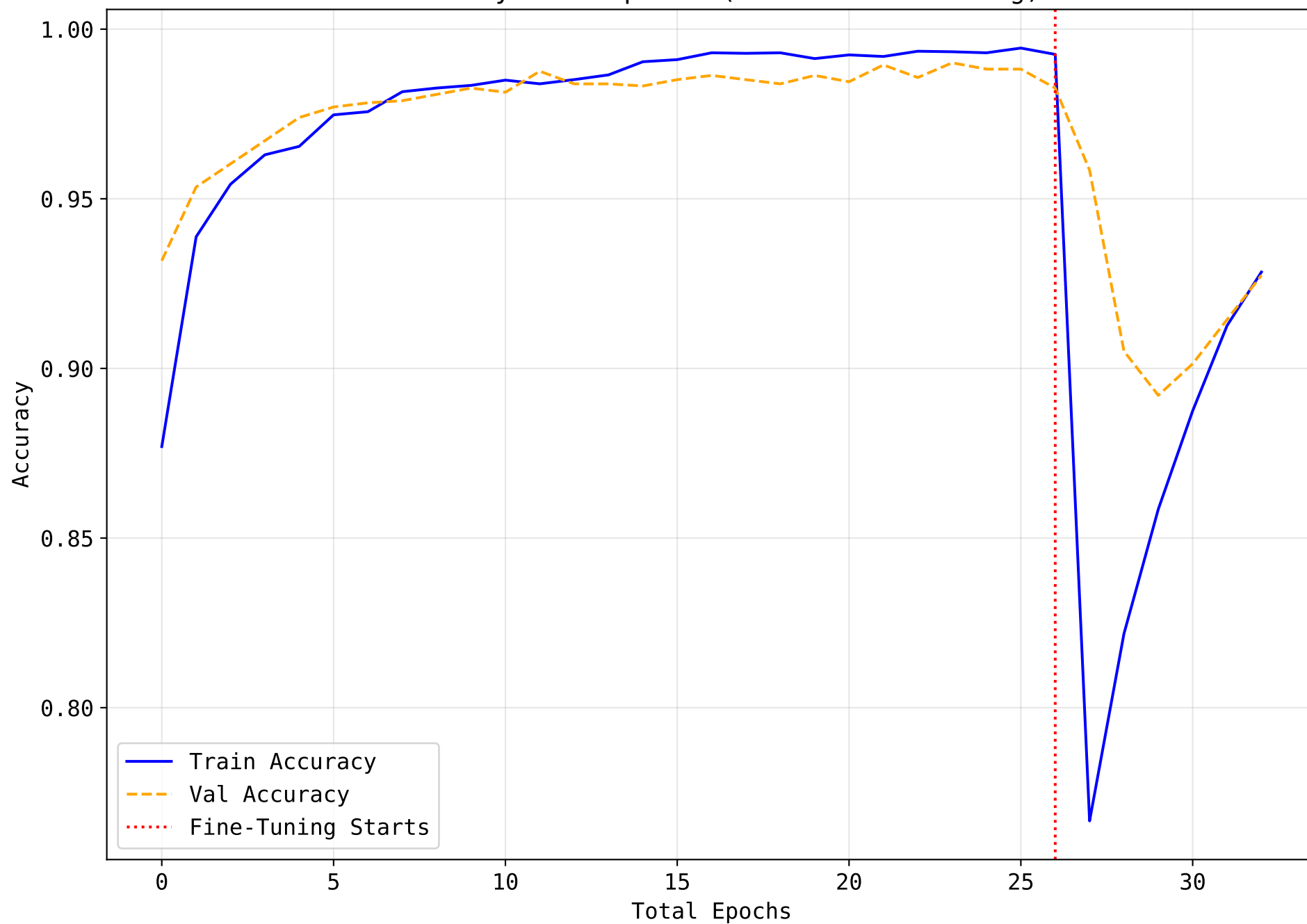
Test Loss : 1.0610
Binary Accuracy : 0.8189
Precision : 0.7454
Sensitivity (Recall) : 0.9595
AUC : 0.9040
Specificity: 0.6830

Optimal Threshold found: 0.9999
Optimised sensitivity: 0.8412
Optimised specificity:0.9314

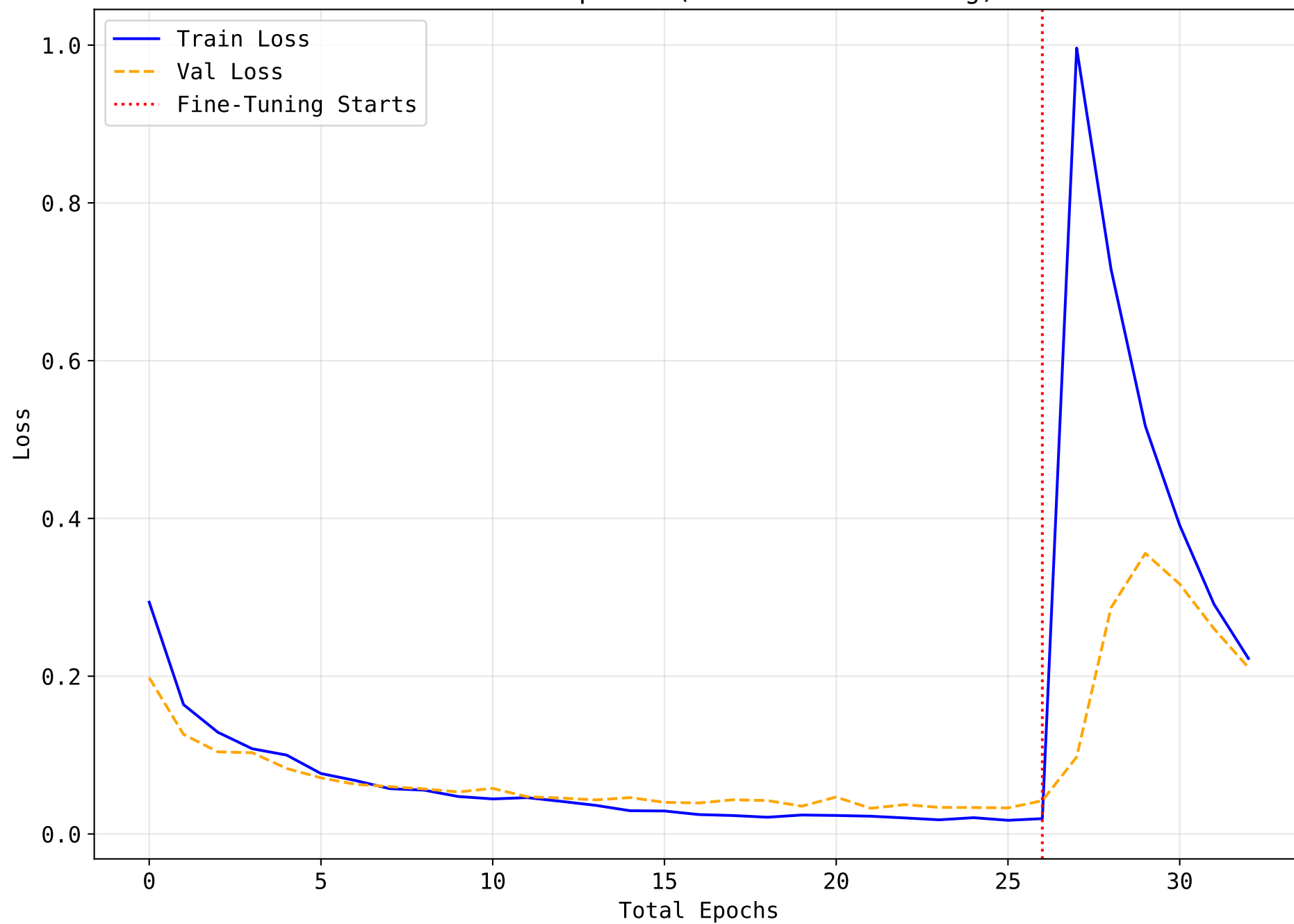
Classification Report:

	precision	recall	f1-score	support
Normal	0.95	0.68	0.79	306
Cataract	0.75	0.96	0.84	296
accuracy			0.82	602
macro avg	0.85	0.82	0.82	602
weighted avg	0.85	0.82	0.82	602

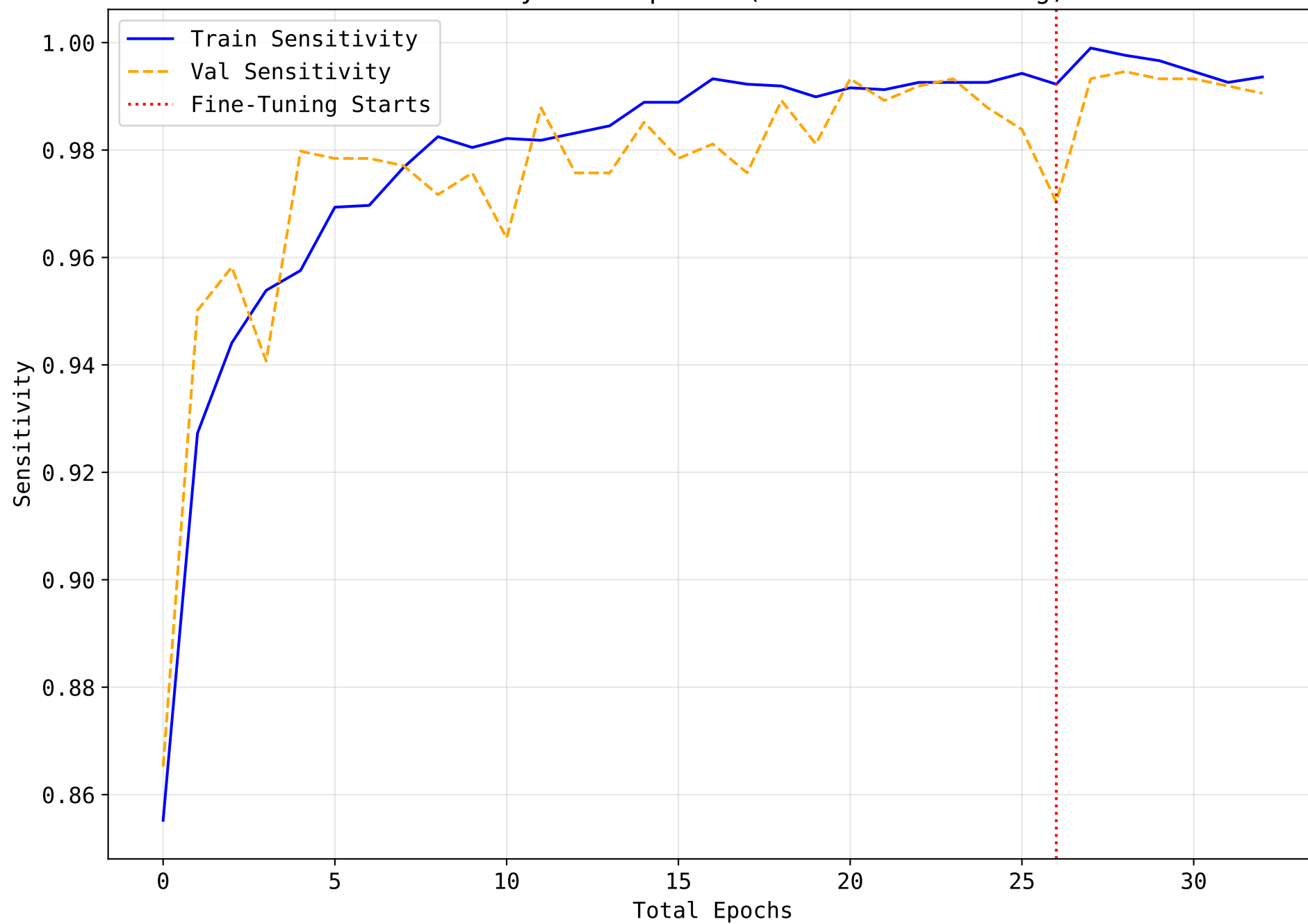
Accuracy Over Epochs (Two-Phase Training)



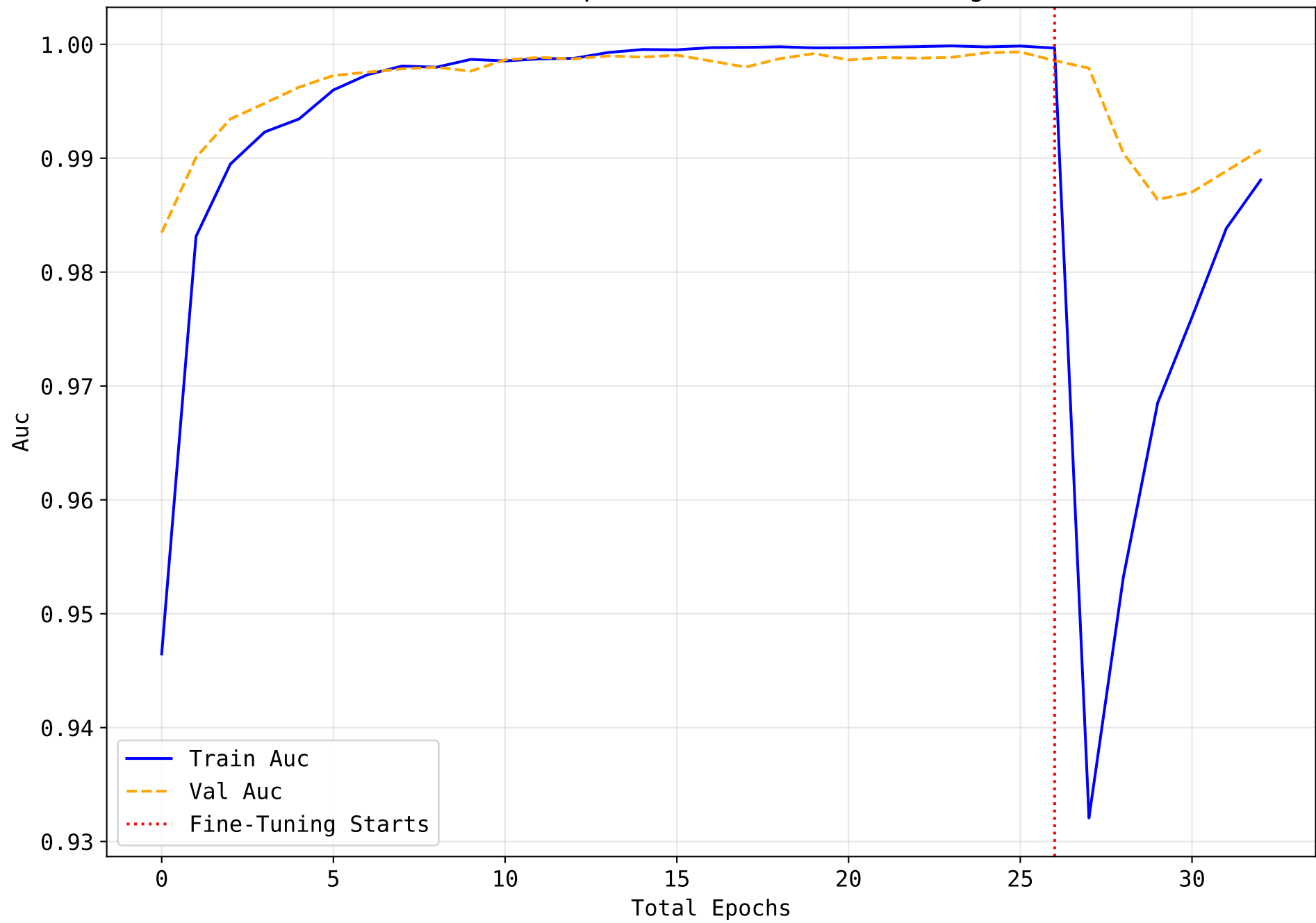
Loss Over Epochs (Two-Phase Training)



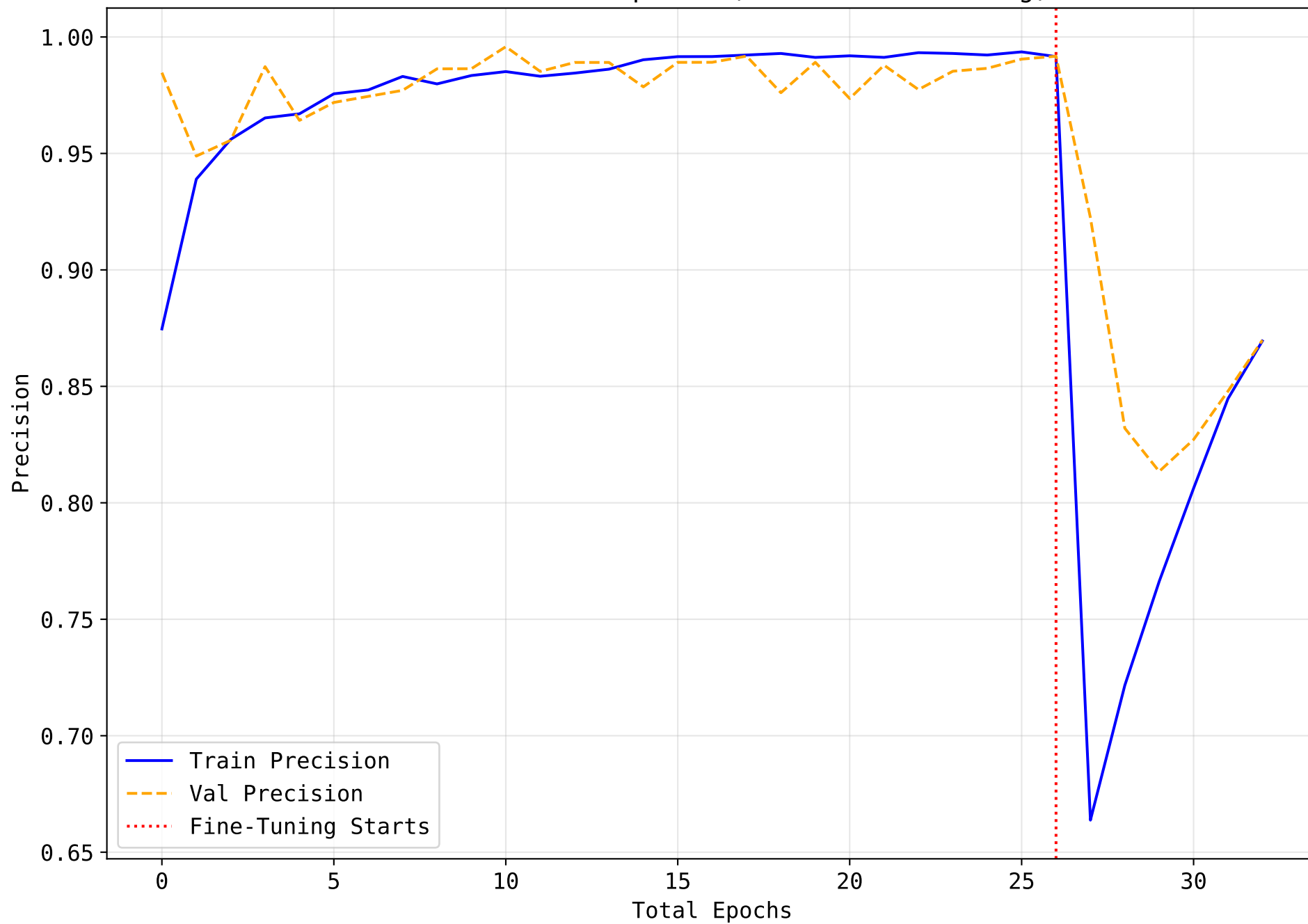
Sensitivity Over Epochs (Two-Phase Training)



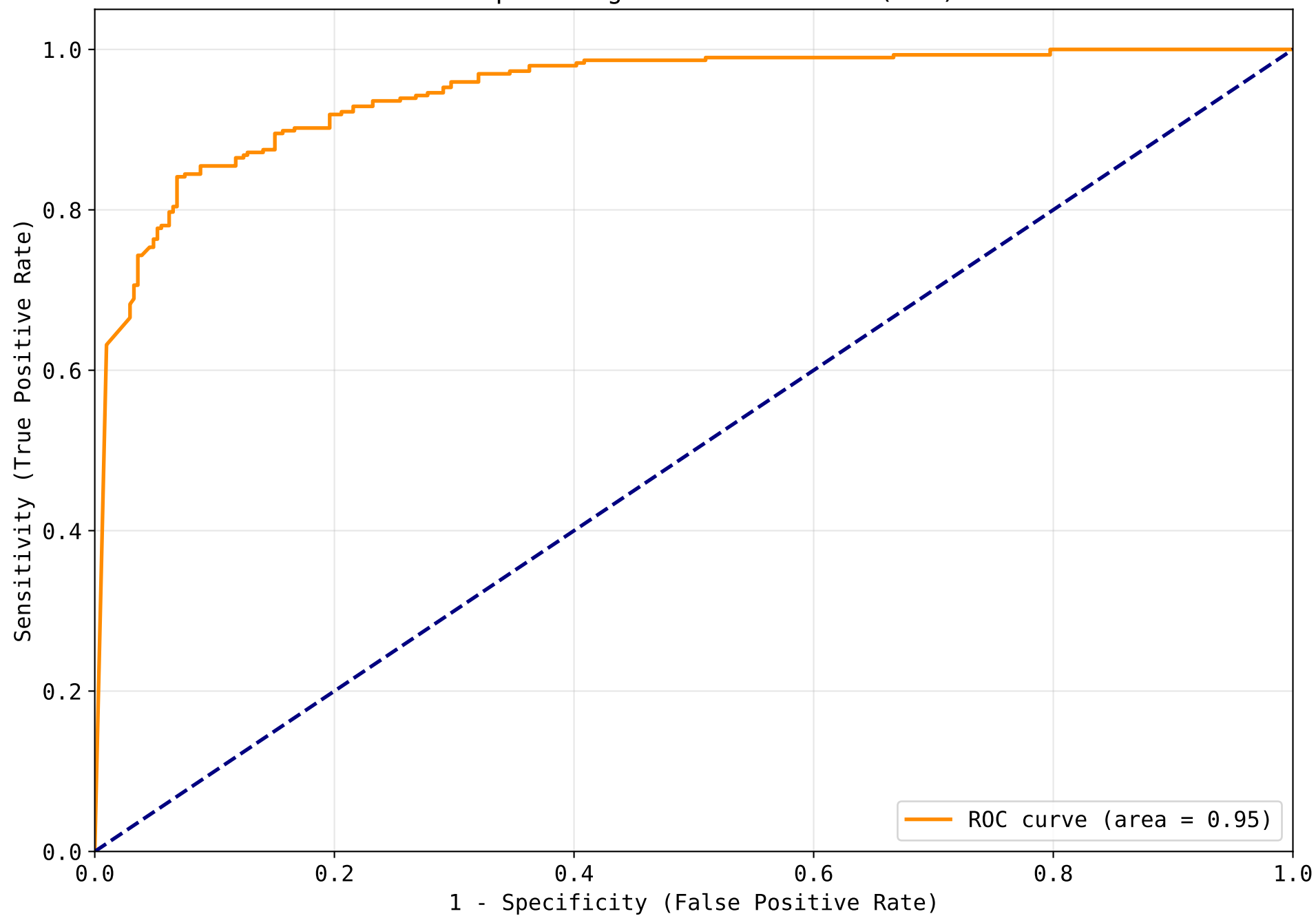
Auc Over Epochs (Two-Phase Training)



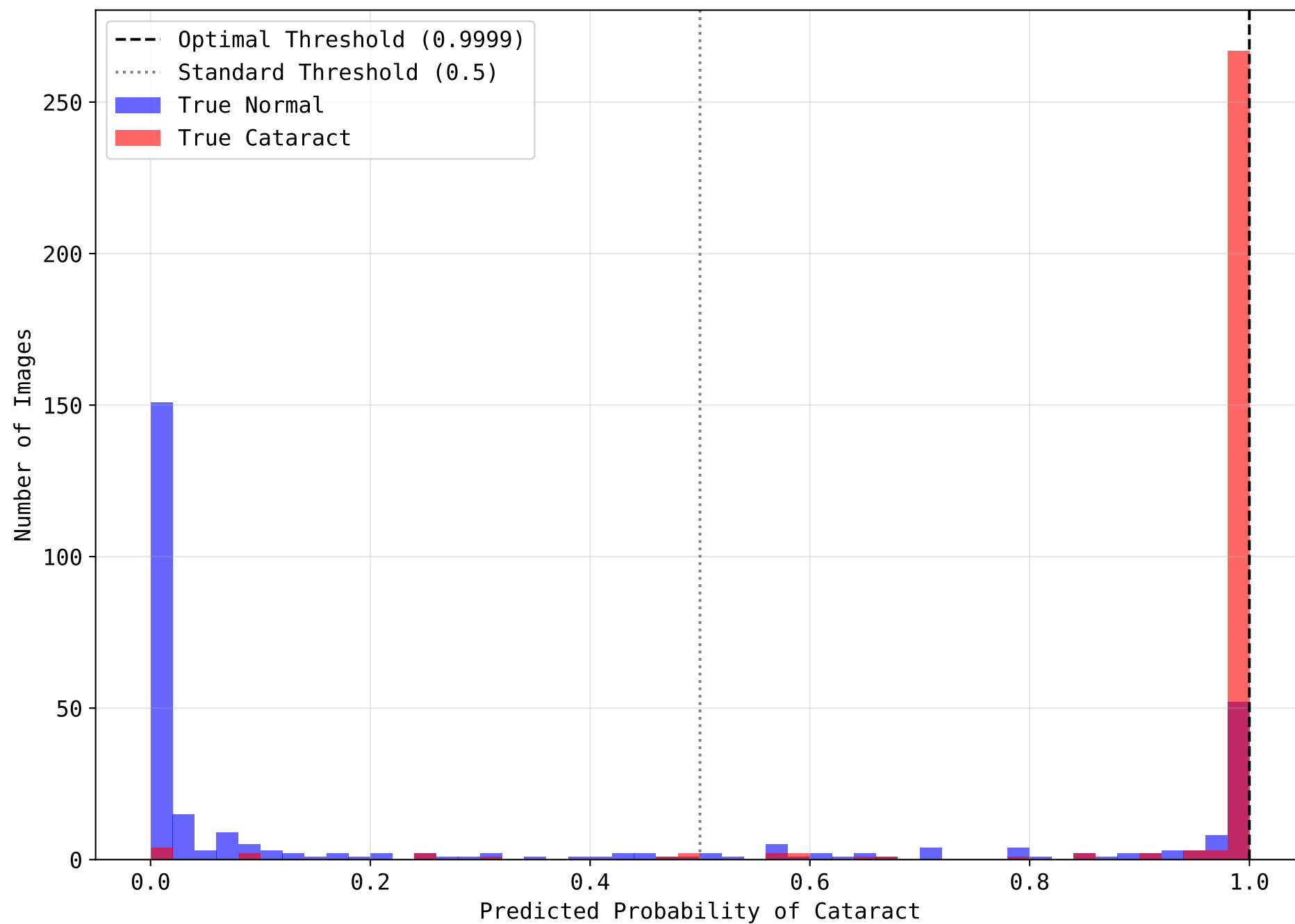
Precision Over Epochs (Two-Phase Training)



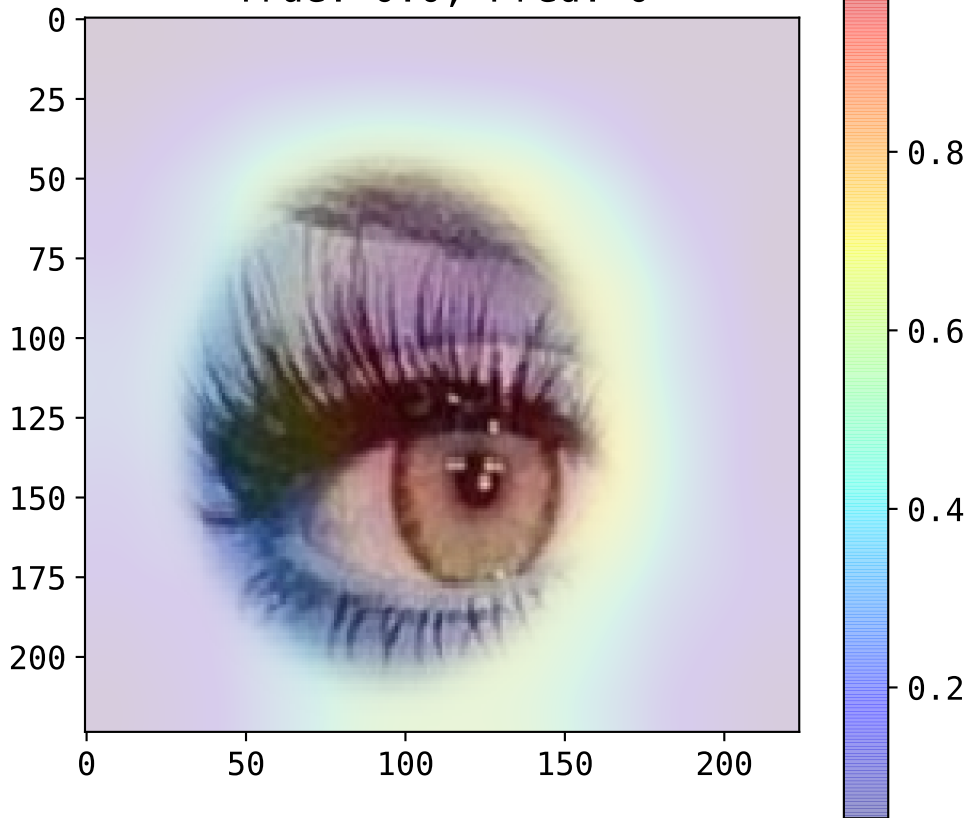
Receiver Operating Characteristic (ROC) Curve



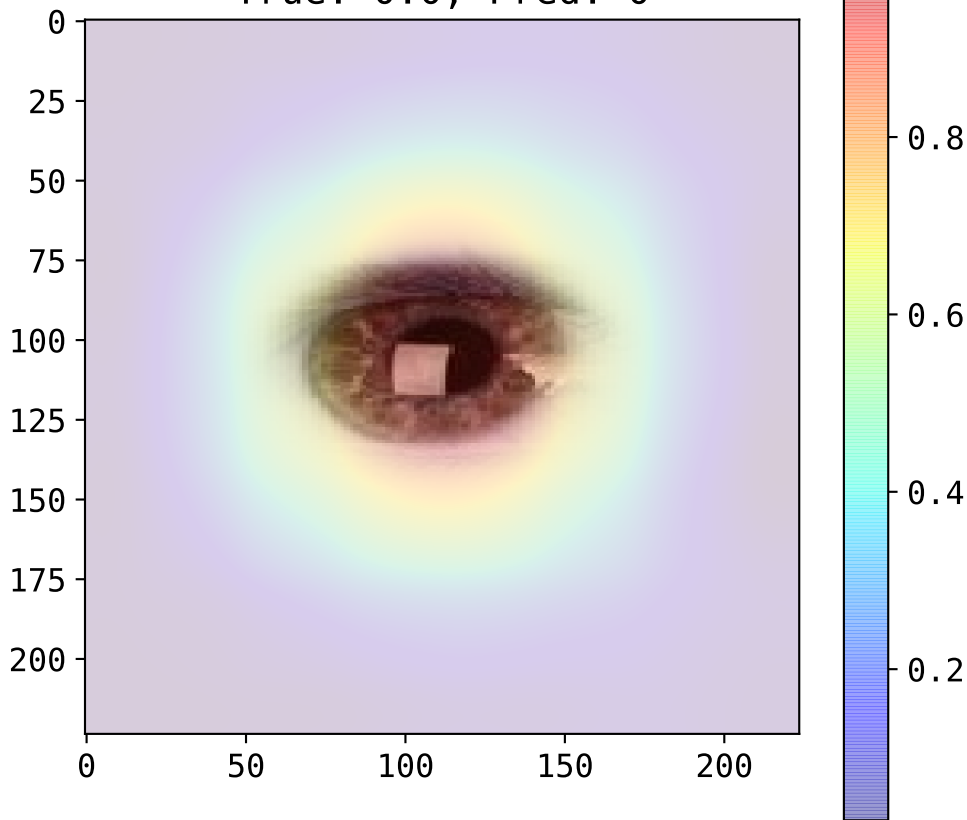
DenseNet121 - Distribution of Predicted Probabilities



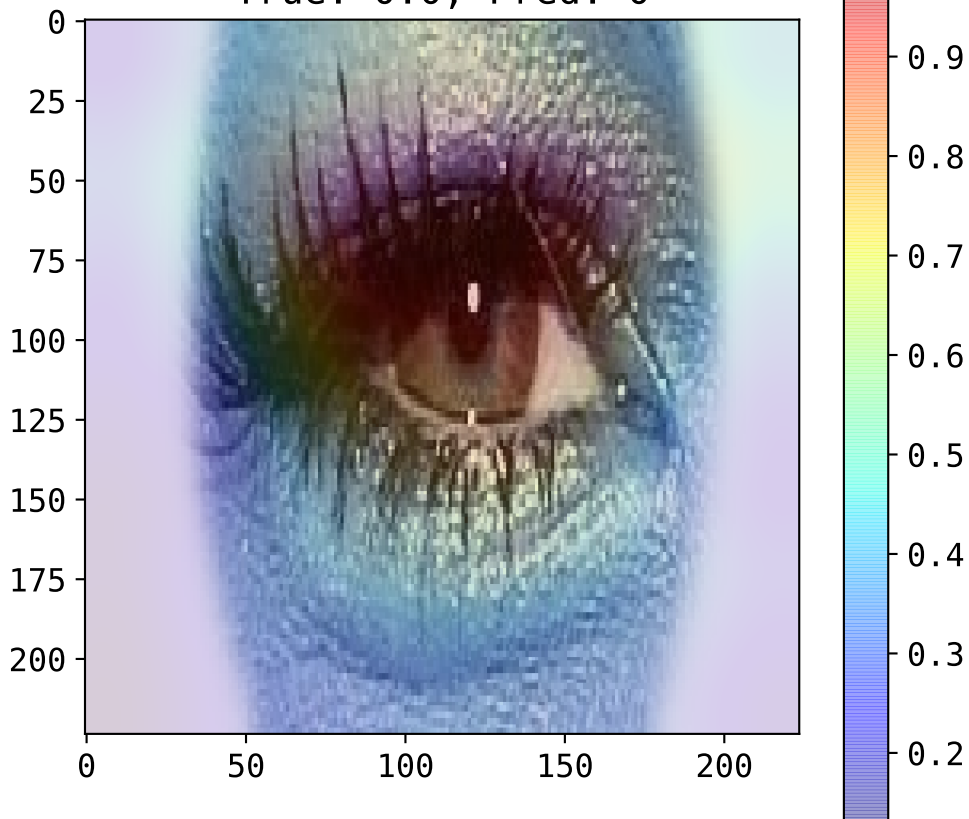
Correctly Classified
True: 0.0, Pred: 0



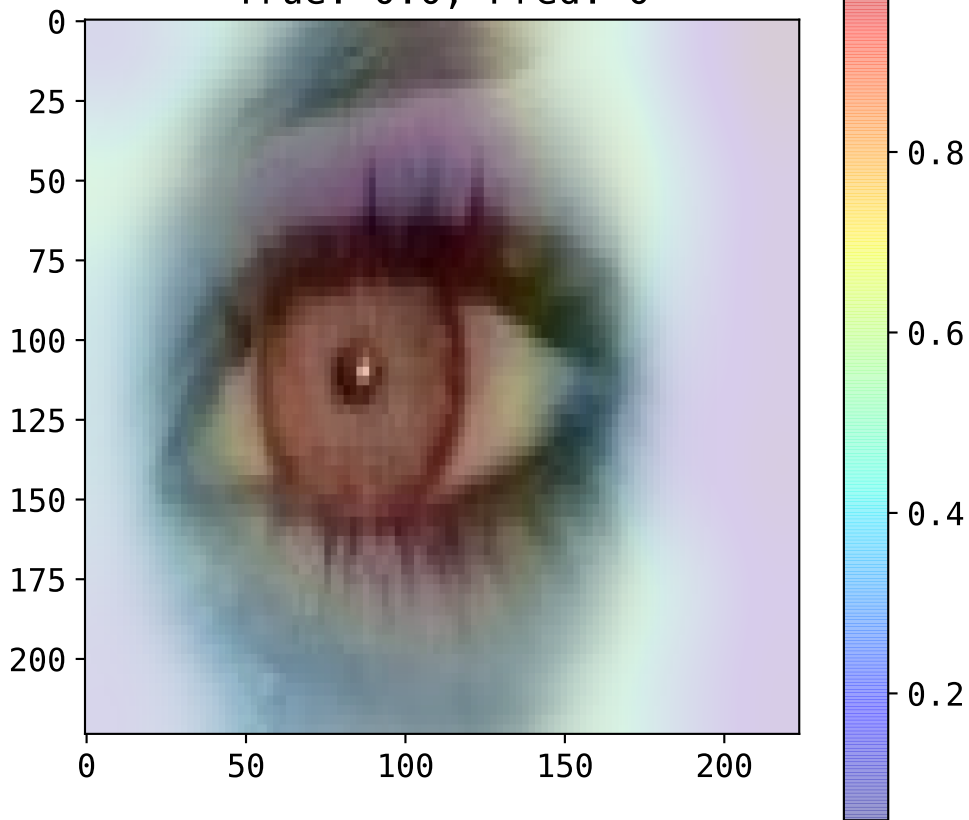
Correctly Classified
True: 0.0, Pred: 0



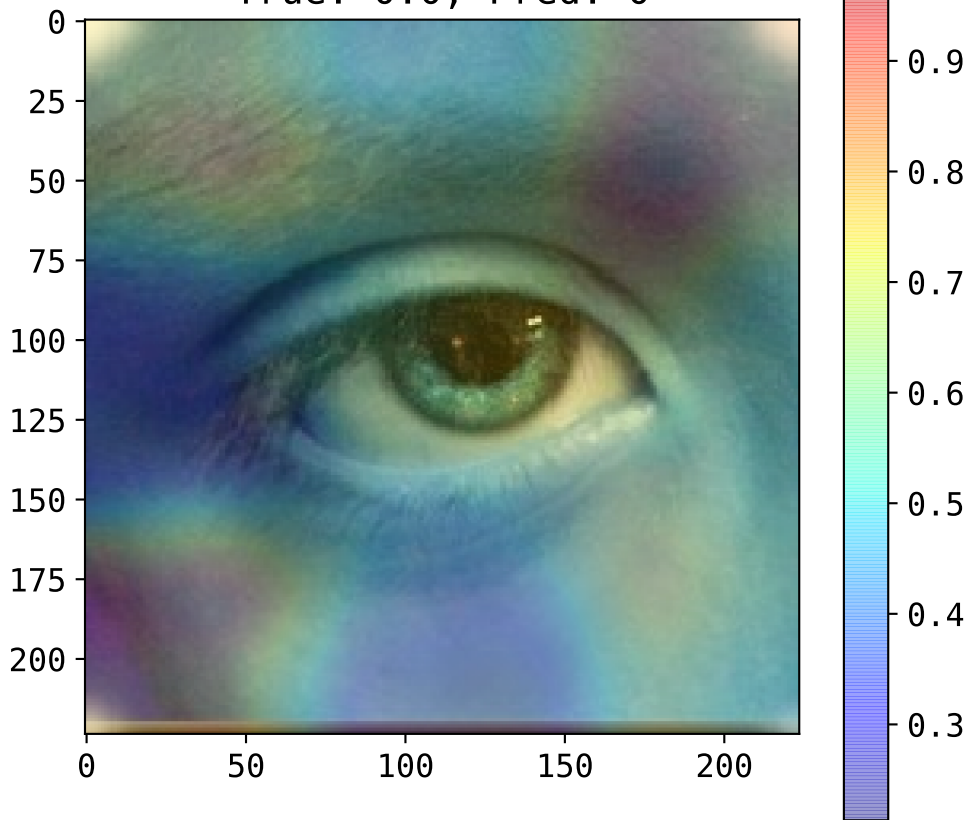
Correctly Classified
True: 0.0, Pred: 0



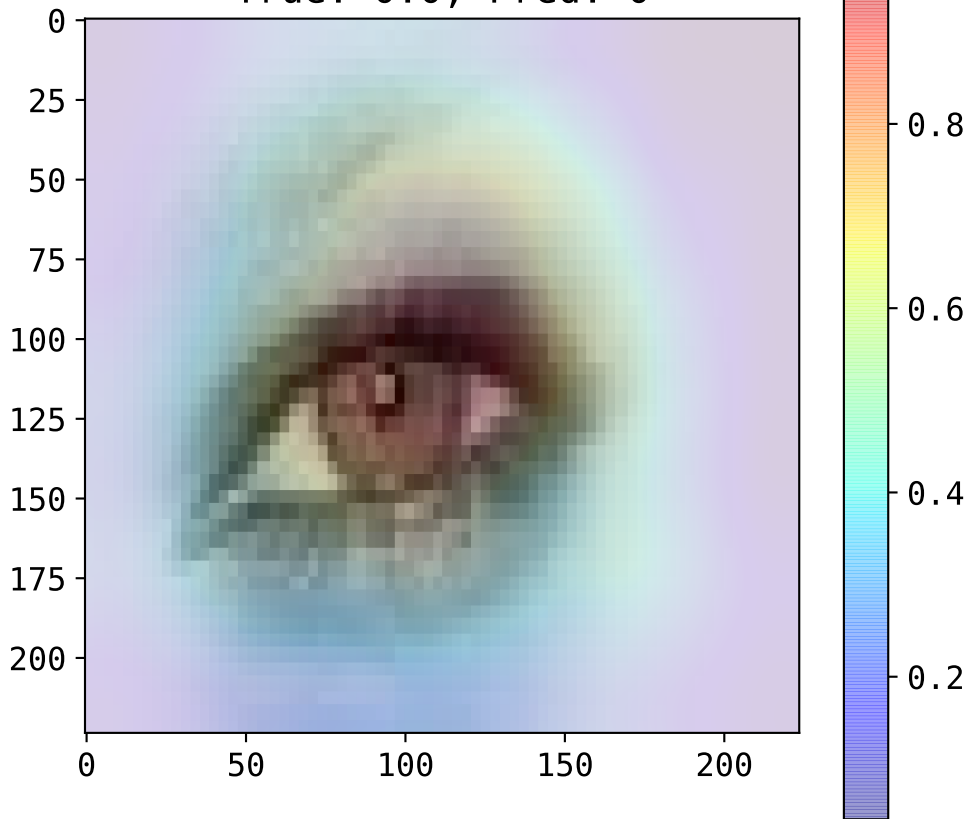
Correctly Classified
True: 0.0, Pred: 0



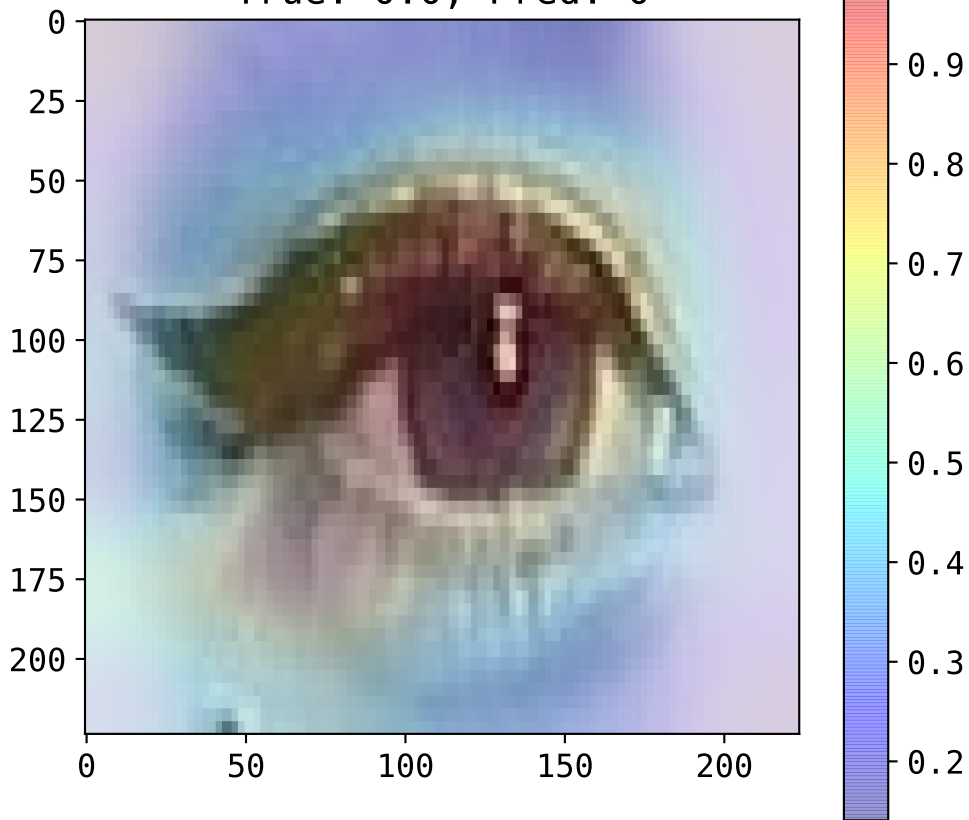
Correctly Classified
True: 0.0, Pred: 0



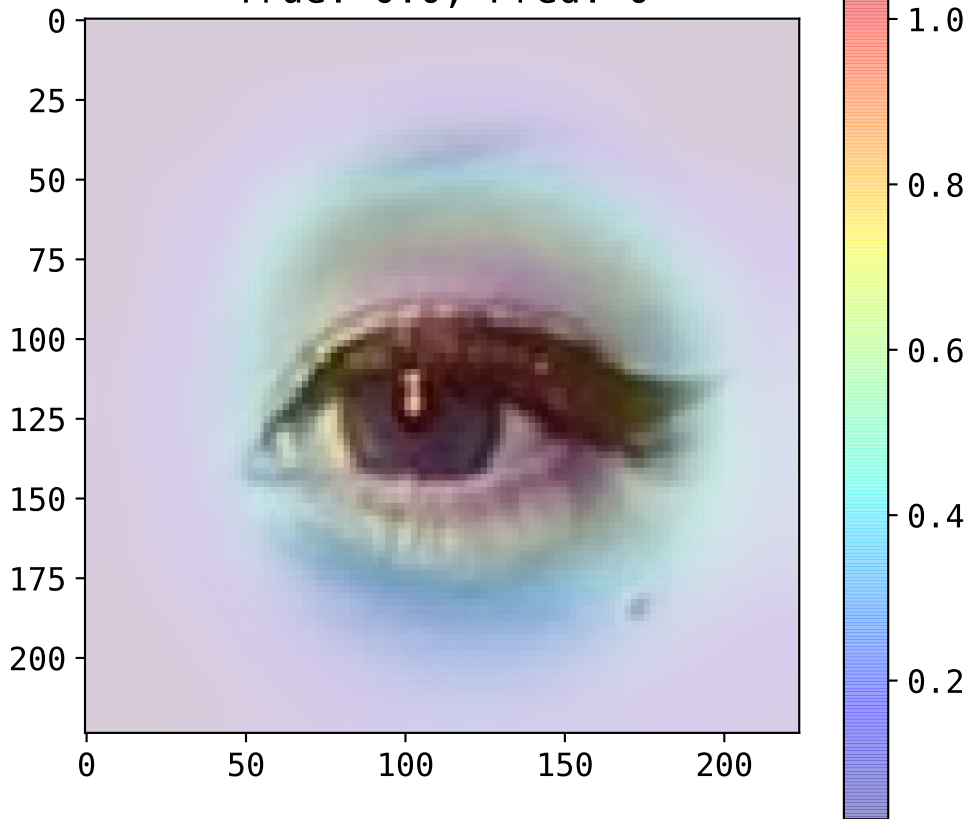
Correctly Classified
True: 0.0, Pred: 0



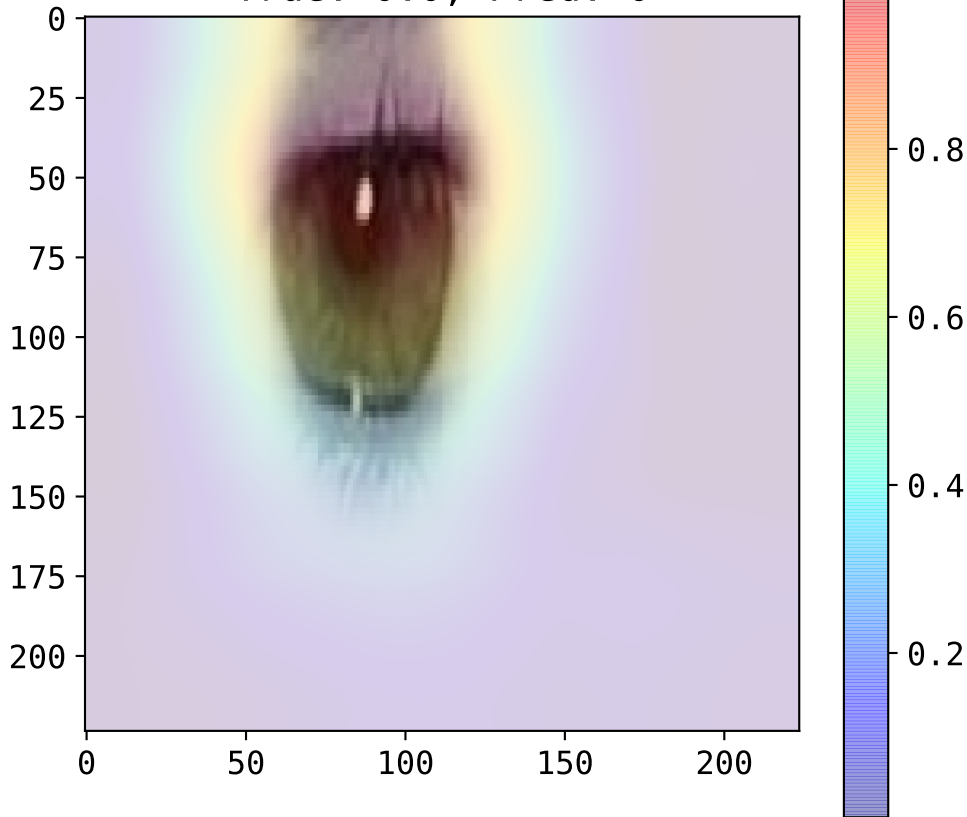
Correctly Classified
True: 0.0, Pred: 0



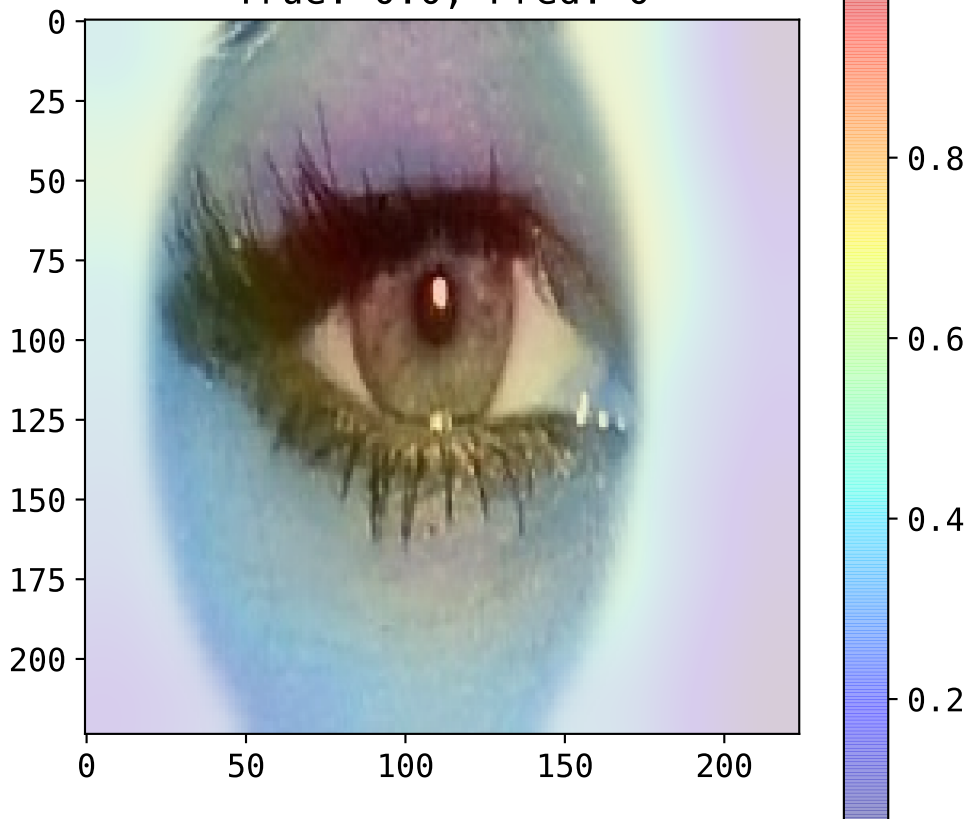
Correctly Classified
True: 0.0, Pred: 0



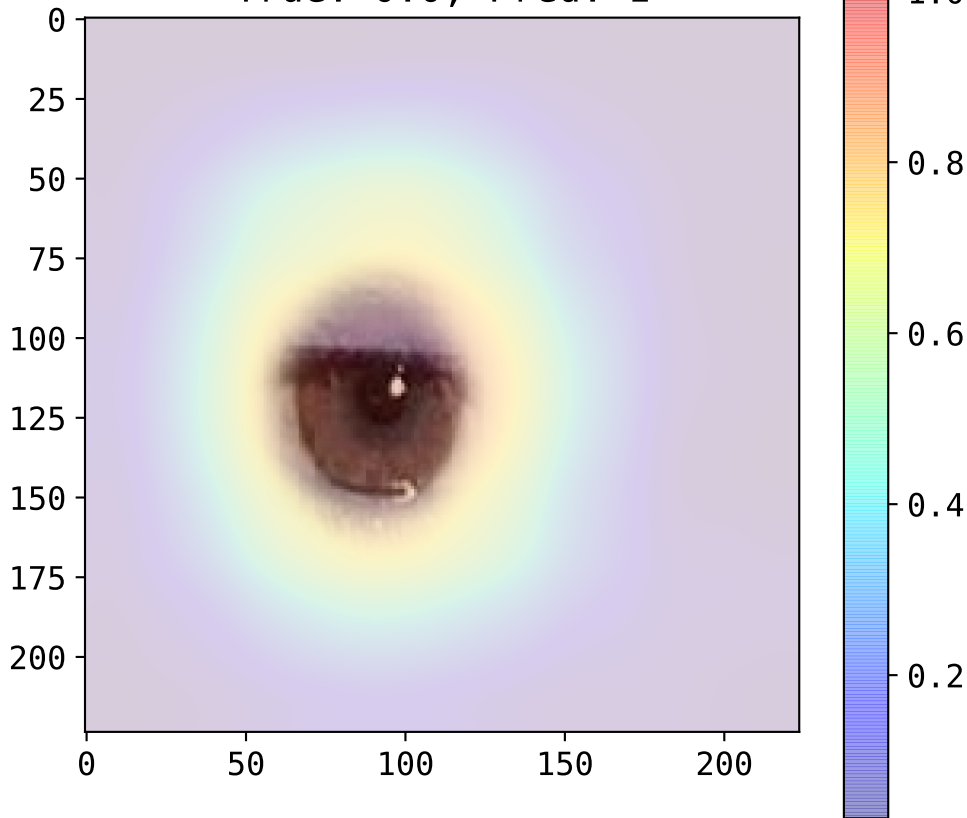
Correctly Classified
True: 0.0, Pred: 0



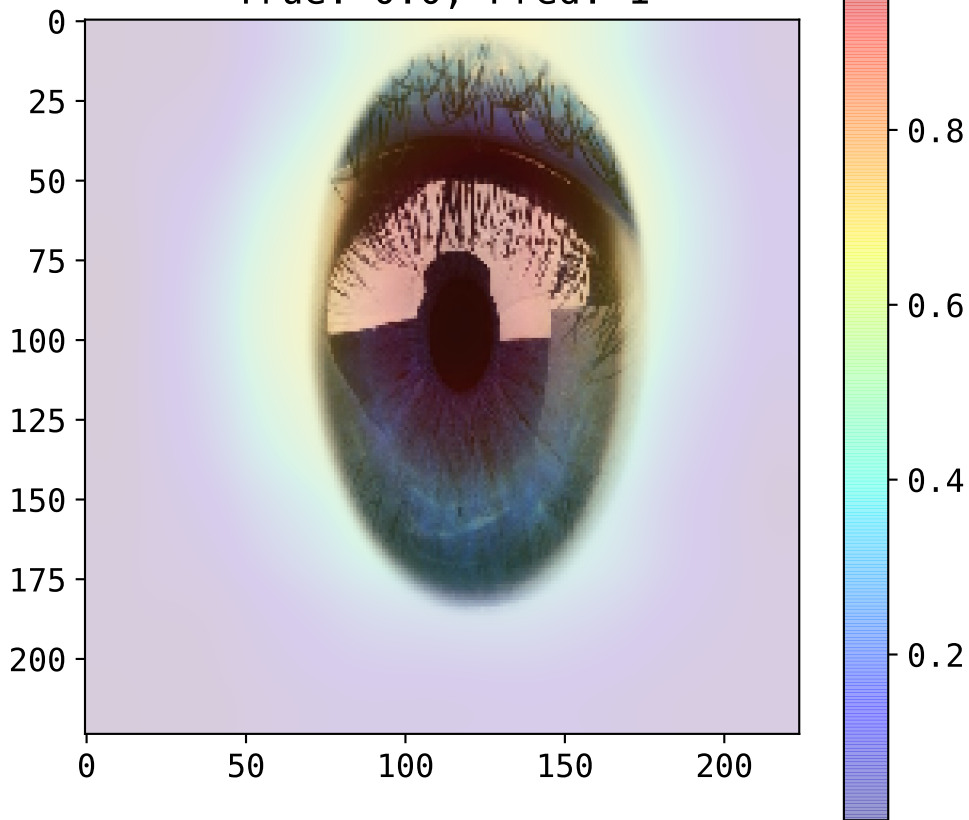
Correctly Classified
True: 0.0, Pred: 0



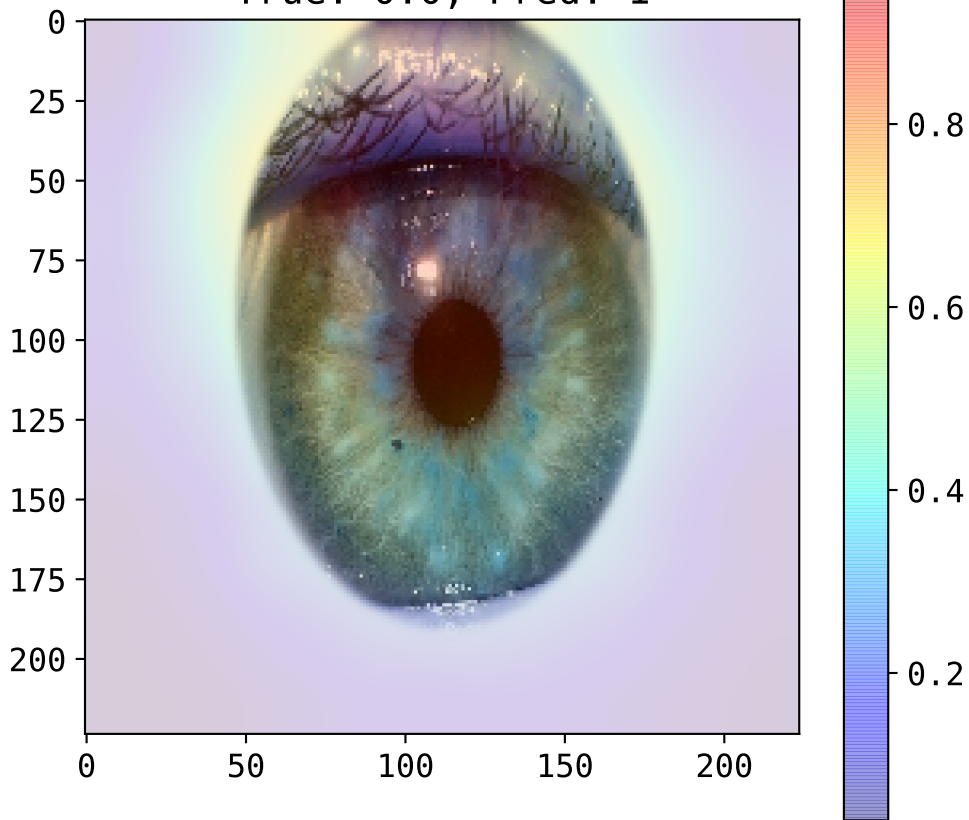
Incorrectly Classified
True: 0.0, Pred: 1



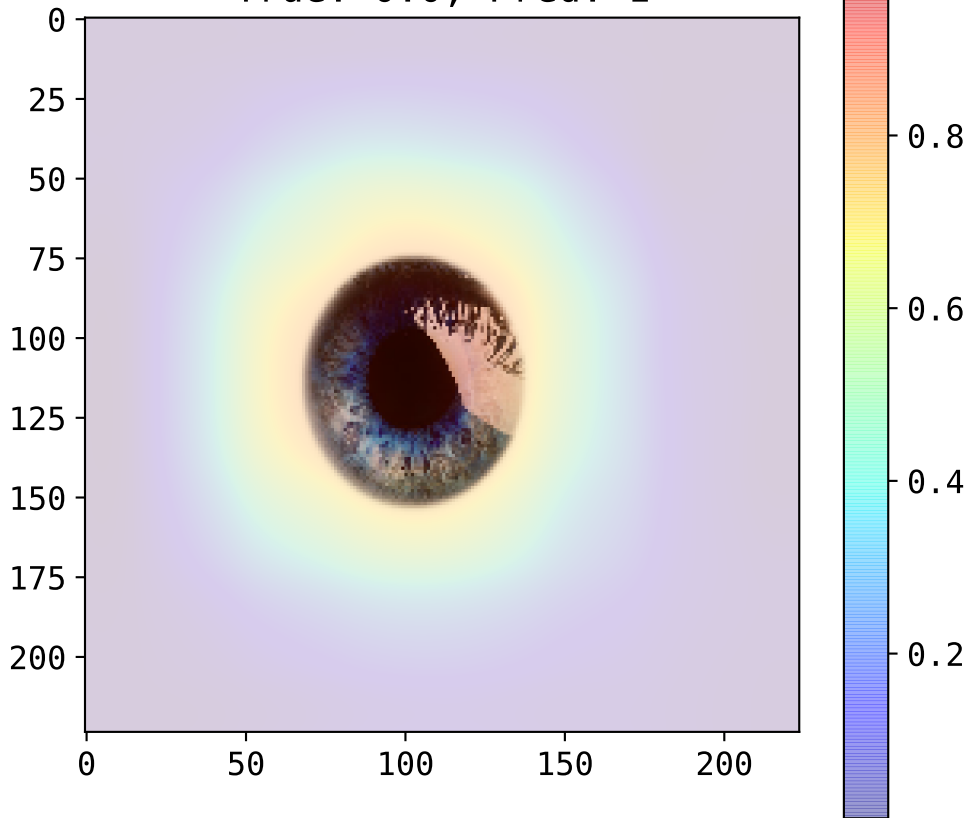
Incorrectly Classified
True: 0.0, Pred: 1



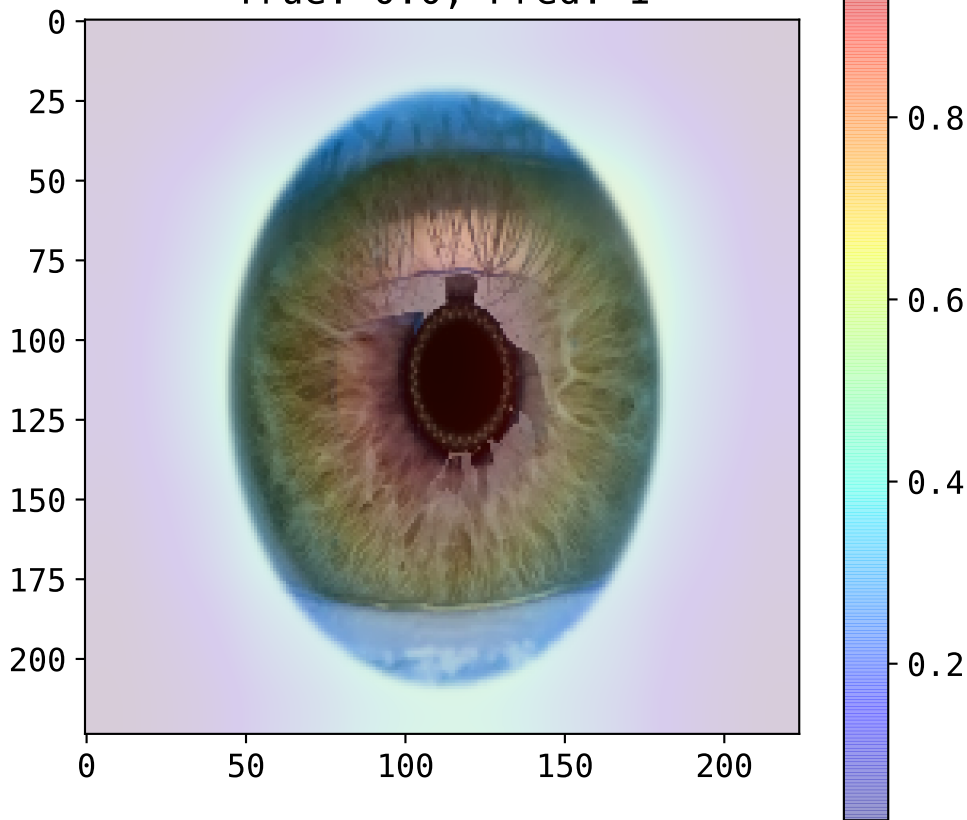
Incorrectly Classified
True: 0.0, Pred: 1



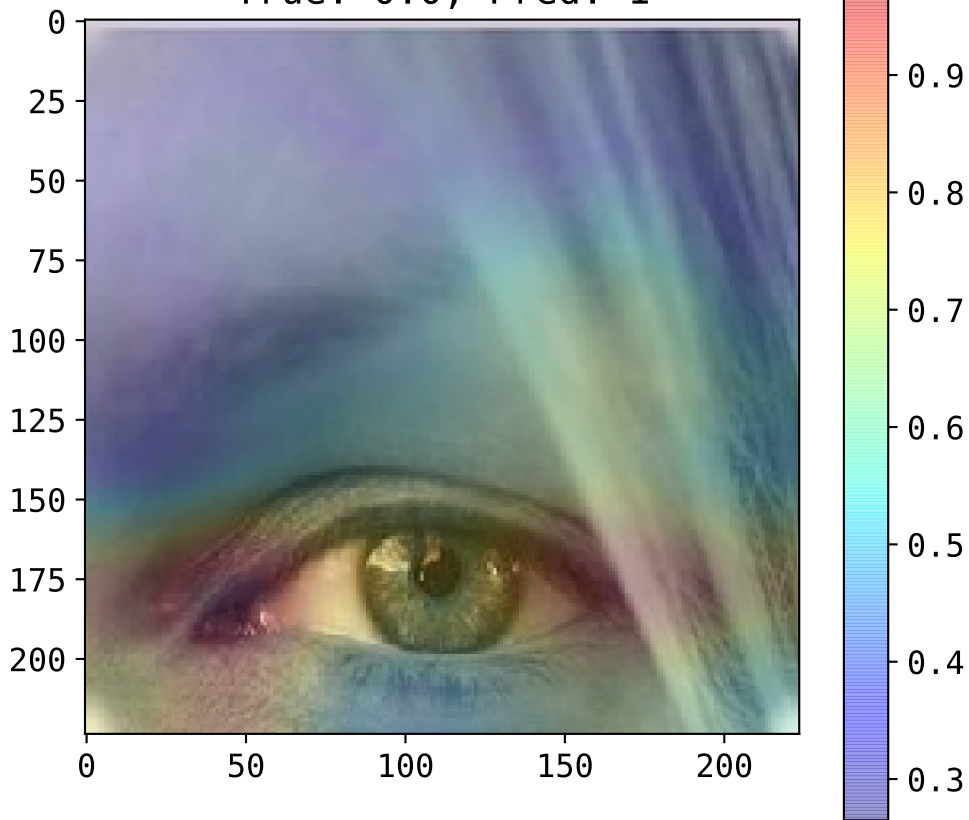
Incorrectly Classified
True: 0.0, Pred: 1



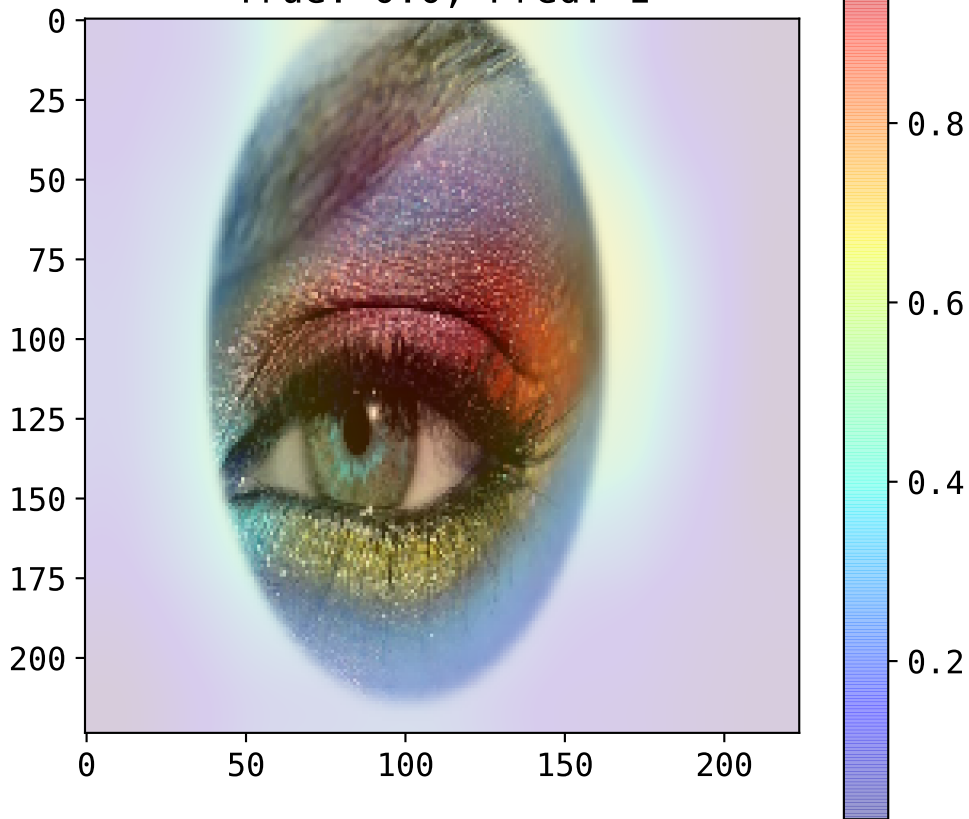
Incorrectly Classified
True: 0.0, Pred: 1



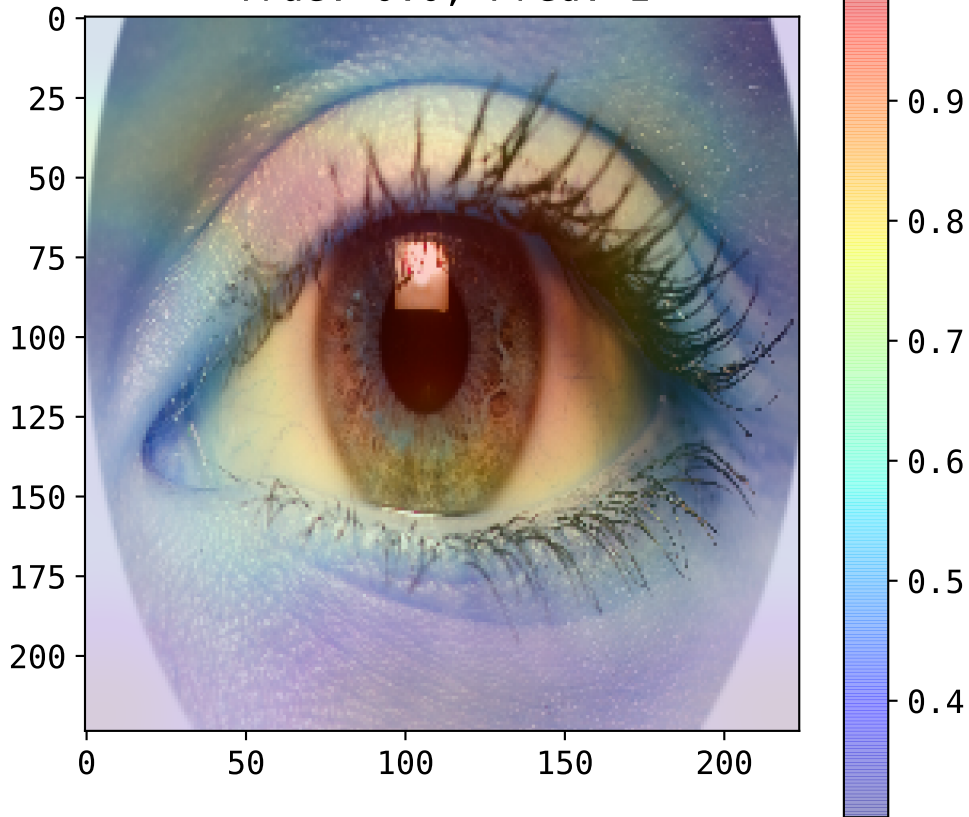
Incorrectly Classified
True: 0.0, Pred: 1



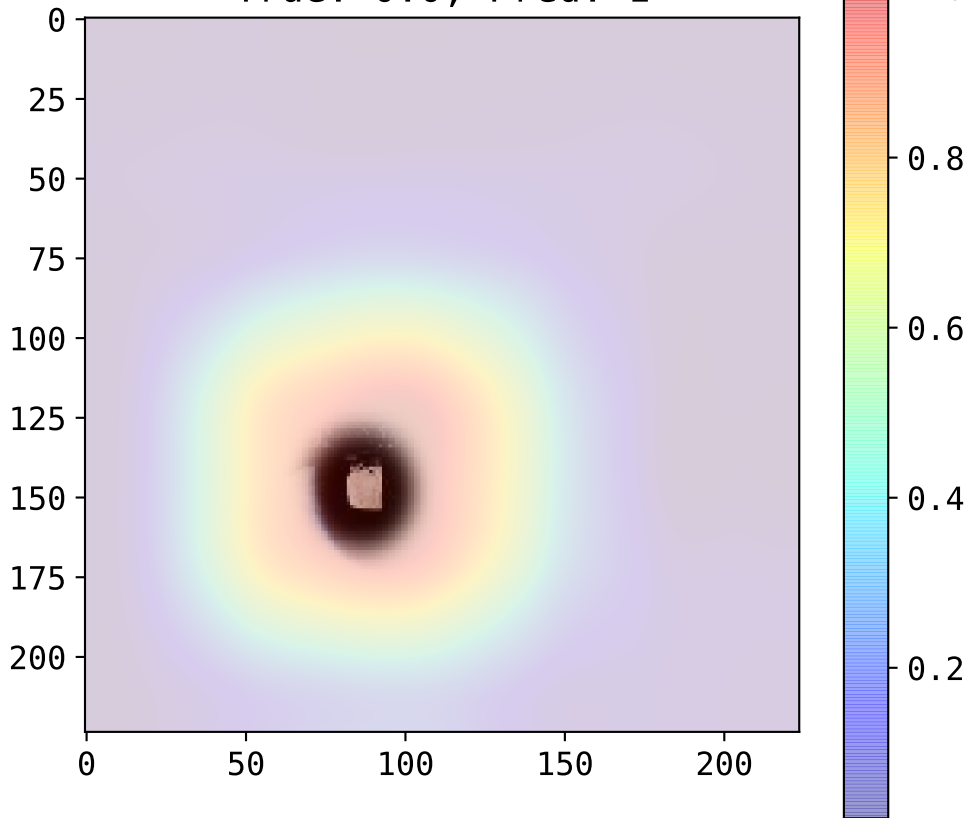
Incorrectly Classified
True: 0.0, Pred: 1



Incorrectly Classified
True: 0.0, Pred: 1



Incorrectly Classified
True: 0.0, Pred: 1



Incorrectly Classified
True: 0.0, Pred: 1

